

Traditional and AI-Generated Scam Text Detection and Risk Scoring Systems

Yu Tao
University of Michigan
Ann Arbor, USA
yutaoa@umich.edu

Abstract—This project presents a unified system for detecting both traditional scam messages and AI-generated scam content. We construct a three-class dataset by combining real call-transcript data with synthetic AI-generated scam texts and train classical machine-learning models using TF-IDF features. Among the evaluated classifiers, Logistic Regression provides the most balanced performance across all classes, achieving strong recall on traditional scams and competitive accuracy on AI-generated scams. We further introduce a probabilistic risk-scoring system that transforms model outputs into an interpretable continuous metric. The resulting risk-score distributions show clear separation among non-scam, traditional scam, and AI-generated scam messages, demonstrating the usefulness of a risk-based assessment. Overall, the system offers a lightweight and effective baseline for scam-text detection and establishes a foundation for future work on emerging AI-assisted fraud.

Index Terms—Scam detection, AI-generated text, TF-IDF features, Logistic Regression, Risk scoring

I. INTRODUCTION

Scam messages have become more complicated in recent years, particularly with the rise of large language models that allow scammers to generate highly polished, formal-looking text. Many individuals, especially those who are unfamiliar with local communication styles, struggle to tell real messages from fraudulent ones. Newcomers frequently encounter suspicious messages in their first semester and often cannot judge whether the content is genuine. This challenge underscores the need for a simple and reliable system that helps users identify potentially harmful messages quickly and with greater confidence.

The goal of this project is to develop a unified classifier capable of detecting both traditional human-written scams and AI-generated scam messages. To accomplish this, the project integrates real phone-call transcripts with synthetic AI-generated texts into a single dataset and applies standard text-processing and machine-learning techniques introduced in class. In addition to classification, it introduces a simple risk-scoring system that provides users with a more interpretable estimate of a message’s likelihood of being fraudulent. Together, these components form a practical early-stage framework for helping users recognize suspicious communications.

Previous research provides important context for this work. Early approaches commonly relied on classical machine-learning models such as Naive Bayes, logistic regression, and SVM with TF-IDF features, which performed well on

simple SMS datasets but often struggled with more complex or evolving scam patterns. More recent studies adopted deep-learning methods. For example, Chen and Chen [1] proposed a BERT-BiLSTM-EAA model that outperformed traditional baselines, demonstrating the advantages of transformer-based architectures. LLM-oriented work has also emerged. Thapa et al. [2] compared classical ML, deep-learning models, and quantized LLMs, showing that lightweight LLMs can reach higher accuracy with reasonable computational cost. Explainability has become another focus. Lim et al. [3] developed EXPLICATE, an ML+LLM hybrid system that combines LIME/SHAP explanations with natural-language summaries and achieves 98.4% accuracy. Jiang [4] further explored GPT-based detection pipelines for phishing and other online fraud. Together, these studies demonstrate strong progress in deep-learning and LLM-based scam detection.

However, most existing studies treat scam detection as a binary classification problem and rarely distinguish AI-generated scams or provide risk-based interpretability, leaving space for a unified system such as the one developed in this project.

II. METHOD

A. Problem Formulation

The goal of this project is to detect and characterize phone-based scam messages, with a particular focus on distinguishing AI-generated scams from traditional human-written ones. Let x denote a short text dialogue, typically representing a brief exchange or conversational. For each x , the learning task is to predict a discrete label

$$y \in \{0, 1, 2\}, \quad (1)$$

where $y = 0$ corresponds to non-scam, $y = 1$ to traditional scam, and $y = 2$ to AI-generated scam.

In addition to this three-way classification, we also define a continuous scam risk score for each message, which is later used to derive low, medium, and high risk categories. This score is computed from the model’s predicted class probabilities and is described in detail in Subsection II-E.

B. Dataset Construction

The dataset used in this work combines real-world traditional scam transcripts with synthetically generated AI-style scam messages.

TABLE I
DATASET CATEGORY STATISTICS

Category	Count
Traditional Scam	397
Non-Scam	253
AI Scam	50
Total	700

TABLE II
FIELDS INCLUDED IN EACH SAMPLE

Field	Description
Id	Unique integer identifier
text	Full call transcript or generated scam message
label_id	Numerical label (0 = non-scam, 1 = traditional scam, 2 = AI-generated scam)
label_text	Readable label
Source	Dataset origin ("Kaggle" or "qwen_generated")
is_synthetic	0 for real data, 1 for AI-generated

1) *Traditional and Non-Scam Data*: The traditional and non-scam samples are sourced from the publicly available *Call Transcripts Scam Determinations* dataset on Kaggle. This dataset contains 60 annotated phone-call transcripts, evenly divided between legitimate and scam calls. Each transcript includes a `CALL_ID`, the full `CALL_TRANSCRIPTS` with speaker turns, and a binary `IS_SCAM_VERDICT` label indicating whether the call is malicious (1) or legitimate (0).

For this project, the original binary labels are mapped to two categories in our unified label space: non-scam (class 0) and traditional scam (class 1). These samples provide real conversational data that help the model learn the typical words used in real versus scam phone interactions.

2) *AI-Generated Scam Data*: To model modern AI-assisted fraud, we construct a synthetic dataset using the Qwen2.5-1.5B-Instruct language model. We design a set of realistic scam scenarios such as AI voice deepfake impersonation, family-emergency scams, and bank security alerts. For each scenario, we prompt the model to generate multiple telephone-style scam messages. All generated samples are labeled as class 2 (AI-generated scam).

3) *Final Dataset and Split*: After cleaning and merging, the combined dataset contains several hundred traditional/non-scam messages and a smaller but distinct subset of AI-generated scams. We perform a stratified split into training, validation, and test sets with proportions 70%, 15%, and 15%, respectively. In our experiments, this results in 490 training samples, 105 validation samples, and 105 test samples, while preserving the class distribution in each split.

C. Text Representation

All models operate on a bag-of-words style representation built with term frequency-inverse document frequency (TF-IDF). Each message is lowercased and tokenized into words. We then apply a `TfidfVectorizer` with an n-gram range of (1, 2), which includes both unigrams and bigrams. This choice allows the model to capture not only individual keywords ("security", "verify") but also short phrases that are

highly indicative of scams ("immediate action", "verify account"). Stopwords are not aggressively removed, since certain function words and polite phrases also appear in distinguishing non-scam customer-service style conversations from scripted scam patterns.

D. Classification Models

We evaluate three classical linear classifiers on the TF-IDF features: logistic regression, multinomial Naive Bayes, and a linear support vector machine.

1) *Logistic Regression (LR)*: Our primary model is a multinomial logistic regression with an ℓ_2 penalty trained on the TF-IDF features. We use the default `lbfgs` solver with a maximum of 300 iterations and set `class_weight="balanced"` so that the loss is re-weighted according to the inverse class frequencies, mitigating the impact of the smaller AI-scam class.

The model is fit with `n_jobs=-1` to enable parallel training. Logistic regression scales well to high-dimensional sparse text features, provides interpretable feature weights, and outputs calibrated class probabilities that we later reuse for risk scoring.

2) *Multinomial Naive Bayes (NB)*: Multinomial Naive Bayes is a common baseline model for text classification. It counts how often each word appears and assumes the words are conditionally independent given the class. The model trains very quickly and performs well when certain keywords strongly separate the classes. However, its simple assumptions can struggle with rare classes and more complex language, such as AI-generated scam messages.

3) *Linear Support Vector Machine (SVM)*: We also use a linear SVM classifier trained on the TF-IDF features. SVMs typically work well on high-dimensional text data. However, the standard `LinearSVC` produces only hard class labels instead of probabilities, which makes it less suitable for later use in risk-scoring tasks.

All models are trained on the training split and evaluated on the held-out test set. The validation split is used to monitor performance and ensure that no obvious overfitting occurs.

E. Risk Scoring System

While multi-class classification yields a discrete label for each message, many real-world applications require a graded notion of scam risk. Using the probability outputs of the logistic regression model, we define a continuous scam risk score to represent the sum of the predicted probabilities for traditional and AI-generated scams,

$$\text{RiskScore}(x) = P(y = 1 | x) + P(y = 2 | x). \quad (2)$$

By construction, $\text{RiskScore}(x) \in [0, 1]$.

For interpretability, we further map this score into three risk levels:

- Low risk: $\text{RiskScore}(x) < 0.3$
- Medium risk: $0.3 \leq \text{RiskScore}(x) < 0.7$
- High risk: $\text{RiskScore}(x) \geq 0.7$

These thresholds are chosen so that the three groups appear clear and reasonable on the validation set. With these rules, the risk-scoring system turns the classifier into a simple early-warning tool: it can rank messages by how risky they are and trigger different actions, such as showing a warning to the user or routing high-risk cases for manual review.

III. RESULTS

A. Data Pipeline

The end-to-end data pipeline consists of four main stages. First, all raw transcripts and AI-generated messages are cleaned, segmented into sentence-level units, and merged into a unified dataset with consistent labels. Second, the dataset is split into training (490 samples), validation (105), and test (105) sets using stratified sampling to preserve class proportions. Third, a TF-IDF vectorizer with unigram-bigram features is fitted on the training set, producing a 2,646-dimensional sparse representation for all samples. Finally, the classifier (Logistic Regression with balanced class weights) is trained on the TF-IDF features, validated on the development set, and evaluated on the held-out test set.

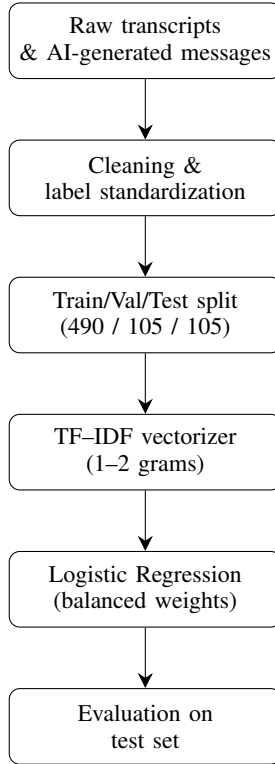


Fig. 1. End-to-end data pipeline for scam-text detection and risk scoring.

B. Classification Performance

1) *Overall Performance*: The classifier demonstrates strong performance on the held-out test set. Using Logistic Regression with TF-IDF features, the model achieves an overall accuracy of 94.29%, with consistently high precision and recall across all three classes (non-scam, traditional scam, and AI-generated scam).

The detailed classification report is shown below:

TABLE III
CLASSIFICATION REPORT ON TEST SET

Class	Precision	Recall	F1-score	Support
Non-scam (0)	0.9714	0.8947	0.9315	38
Traditional scam (1)	0.9219	0.9833	0.9516	60
AI scam (2)	1.0000	0.8571	0.9231	7
Accuracy		0.9429		105
Macro Avg	0.9644	0.9117	0.9354	105
Weighted Avg	0.9450	0.9429	0.9424	105

These results highlight that traditional scam messages (class 1) are the easiest to detect, with recall above 0.98. Non-scam messages show slightly lower recall due to occasional false positives. Despite having only 7 examples, AI-generated scams still achieve high precision (1.00) and competitive recall (0.857), indicating good model sensitivity to synthetic scam patterns.

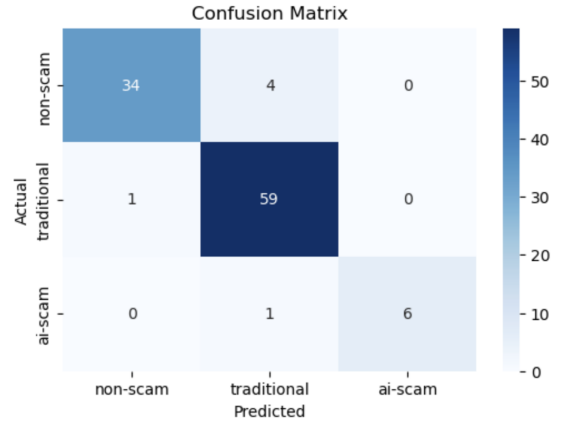


Fig. 2. Confusion Matrix on Test Set

2) *Confusion Matrix Analysis*: The confusion matrix shows that the model distinguishes the three classes reliably. Non-scam messages are mostly identified correctly (34/38), with a small number misclassified as traditional scams. Traditional scams achieve strong separation (59/60 correct), indicating that the model captures typical scam-related cues. For AI-generated scams, 6 out of 7 samples are correctly classified, with only one confusion with traditional scams. Importantly, no AI-generated scam is predicted as non-scam, suggesting low false-negative risk.

C. Risk Scoring Analysis

To assess how effectively the model separates legitimate and fraudulent content, we analyze both the density distribution and boxplot of risk scores, revealing class-specific trends.

1) *Risk Score Distribution (KDE)*: The kernel density curves in Fig. 3 show three largely separated risk score regions:

- Non-scam messages cluster around 0–0.4, forming a low-risk distribution.
- Traditional scams mainly fall between 0.6–0.9, indicating strong scam indicators.

- AI-generated scams exhibit the highest scores, typically above 0.9.

Although these distributions show minimal overlap, the class-specific peaks remain clearly distinct, demonstrating that the risk-scoring system effectively differentiates the three message types.

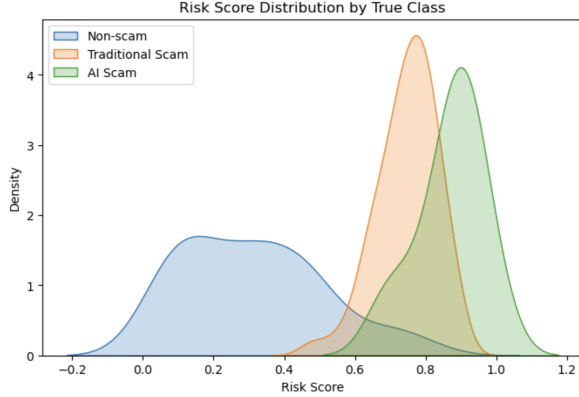


Fig. 3. Risk Score Distribution by True Class (KDE)

2) *Risk Score by Class (Boxplot)*: The boxplot in Fig. 4 further highlights differences in score ranges:

- Non-scam messages have the lowest median and widest variance.
- Traditional scams show a tighter, higher-risk band centered near 0.75.
- AI-generated scams have the highest median and most compact range.

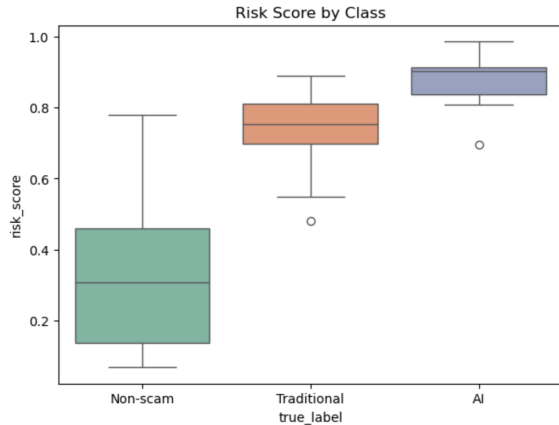


Fig. 4. Risk Score Distribution by Class (Boxplot)

3) *Model Comparison*: To understand how the three classifiers behave on our combined scam-detection dataset, we evaluate Logistic Regression (LR), Naive Bayes (NB), and Linear SVM using the same TF-IDF features and the same train/validation/test splits. This setup allows us to compare how differences in model assumptions affect their performance. A summary of the results is presented in Table IV.

Among the three classifiers, Logistic Regression shows the most balanced and reliable performance. Naive Bayes achieves

TABLE IV
COMPARISON OF CLASSIFICATION PERFORMANCE ACROSS MODELS

Metric	LR	NB	SVM
Accuracy	0.9429	0.8857	0.9238
Macro Recall	0.9117	0.6617	0.8586
Macro F1	0.9354	0.6911	0.8959
AI-scam Recall	0.8571	0.1429	0.7143
AI-scam F1	0.9231	0.2500	0.8333

high precision but performs poorly on the AI-scam class (recall = 0.1429), indicating that it cannot capture the more varied language used in AI-generated scams. Linear SVM is more stable than Naive Bayes and performs well on non-scams and traditional scam messages, but its recall for AI-scam (0.7143) remains lower than ideal.

Overall, Logistic Regression attains the highest accuracy (94.29%) and exhibits the most even performance across all classes, including strong performance on AI-generated scams. These results show that Logistic Regression is the most dependable baseline for building the risk-scoring system.

IV. CONCLUSION

This project presents a unified framework for detecting both traditional scams and AI-generated scam messages. By integrating real phone-call transcripts with synthetically generated scam texts, applying TF-IDF representations, and evaluating multiple classical classifiers, the system demonstrates strong performance on the three-way classification task. Logistic Regression achieves the most balanced accuracy and recall across all classes.

Beyond discrete classification, the project introduces a simple risk-scoring system that converts model probabilities into an interpretable continuous measure of scam likelihood. The resulting density distributions and boxplots show clear separation between non-scams, traditional scams, and AI-generated scam messages, indicating that the risk score provides meaningful graded assessments suitable for early-warning applications.

Overall, the results demonstrate that combining classical machine-learning methods with synthetic data is an effective approach for addressing emerging AI-assisted fraud. Although the dataset—especially the AI-generated scam portion—remains limited in size, this work provides a practical foundation for future extensions involving larger datasets and more advanced detection models.

REFERENCES

- [1] W. Chen and C. Chen, “Deep learning-based model for detecting fraudulent SMS messages,” in *Proc. ICIEAI '24: 2024 2nd Int. Conf. Information Education and Artificial Intelligence*, pp. 346–350, May 2025, doi: 10.1145/3724504.3724561.
- [2] J. Thapa, G. Chahal, S. V. Gabreanu, and Y. Otoum, “Phishing detection in the Gen-AI era: Quantized LLMs vs classical models,” *arXiv preprint arXiv:2507.07406*, 2025.
- [3] B. Lim, R. Huerta, A. Sotelo, A. Quintela, and P. Kumar, “EXPLICATE: Enhancing phishing detection through explainable AI and LLM-powered interpretability,” *arXiv preprint arXiv:2503.20796*, 2025.
- [4] L. Jiang, “Detecting scams using large language models,” *arXiv preprint arXiv:2402.03147*, 2024.